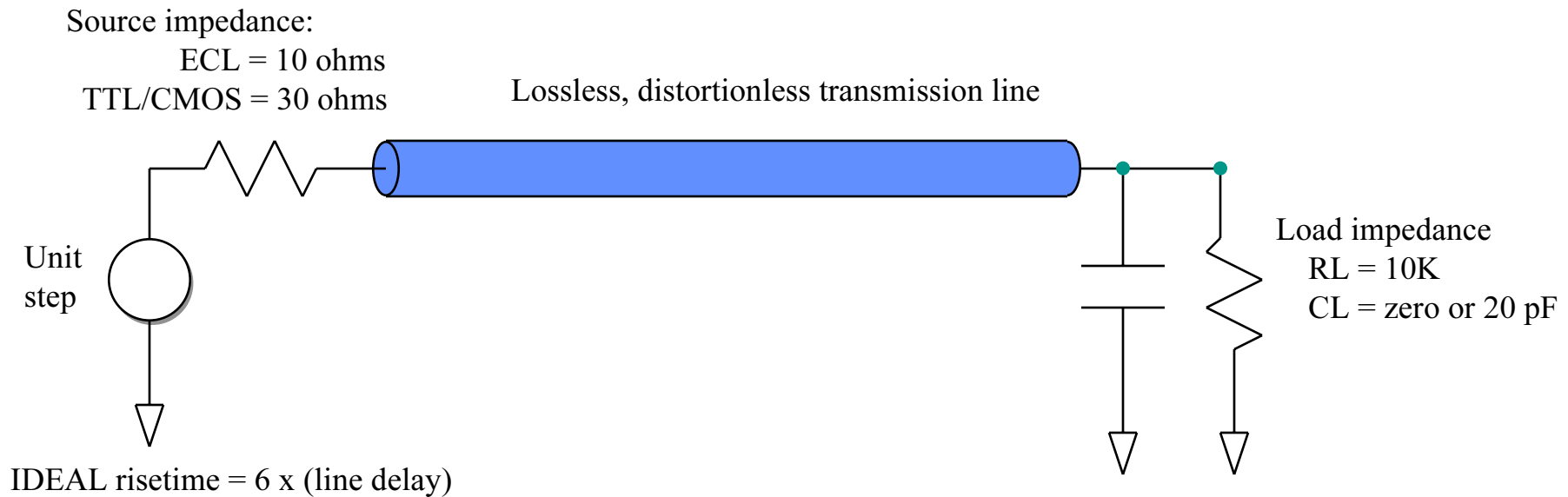


Short Transmission Line Examples



It's the Risetime/Delay Ratio that Counts

Set nominal transmission
line delay and risetime

$$\text{delay} := 10^{-9}$$

$$ZS = 10$$

$$RL = 1 \cdot 10^4$$

$$\text{risetime} := 2 \cdot \text{delay}$$

$$ZC = 65$$

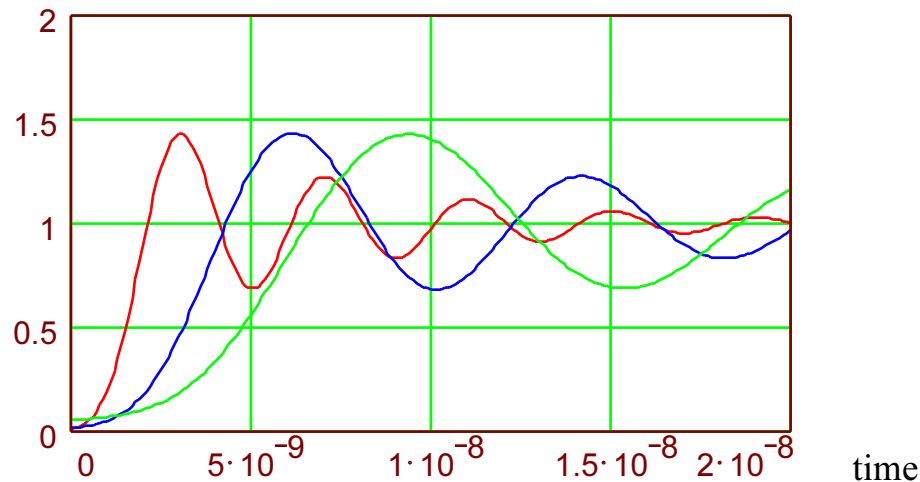
$$CL = 0$$

Scale both delay and risetime to see
what happens

$$X1 := \text{SYS3}(\text{delay}, \text{risetime})$$

$$X2 := \text{SYS3}(\text{delay} \cdot 2, \text{risetime} \cdot 2)$$

$$X3 := \text{SYS3}(\text{delay} \cdot 3, \text{risetime} \cdot 3)$$



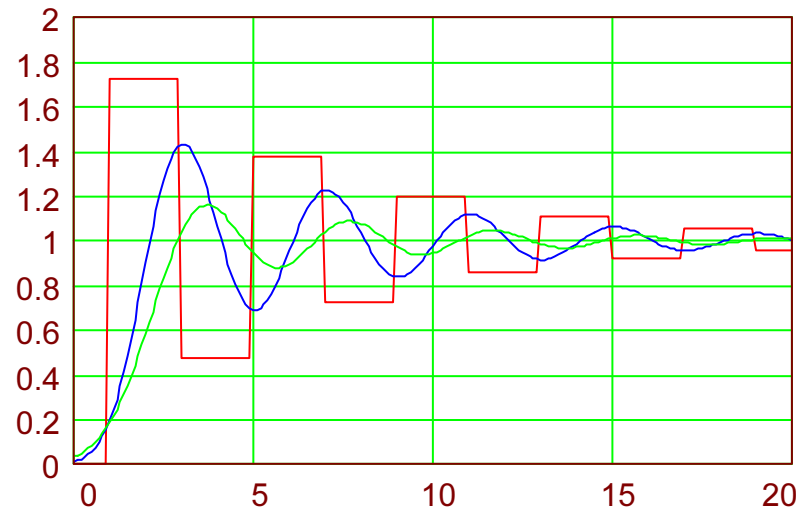
ECL

Unterminated line response

Risetime set to 0, 2 and 3 times transmission line delay

$$Z_S = 10 \quad R_L = 1 \cdot 10^4$$

$$Z_C = 65 \quad C_L = 0$$



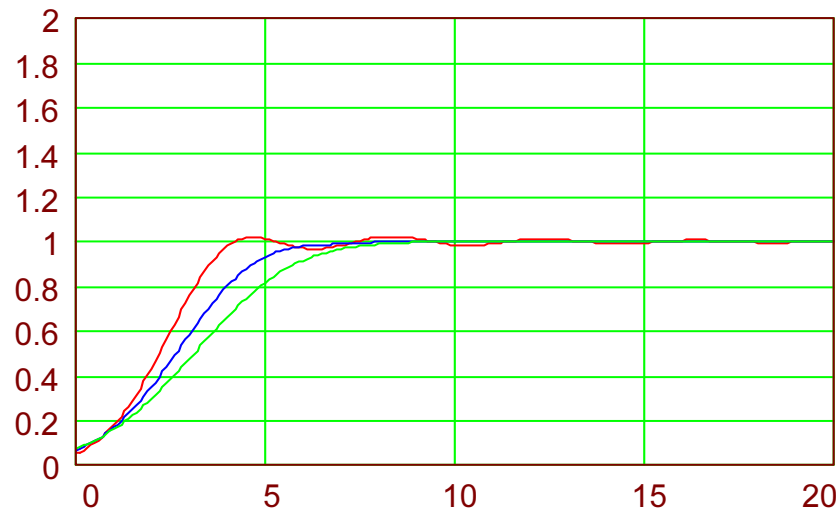
time, in units
of line delay

ECL

Unterminated line response

Risetime set to 4, 5 and 6 times transmission line delay

$$\begin{array}{ll} Z_S = 10 & R_L = 1 \cdot 10^4 \\ Z_C = 65 & C_L = 0 \end{array}$$



time, in units
of line delay

ECL

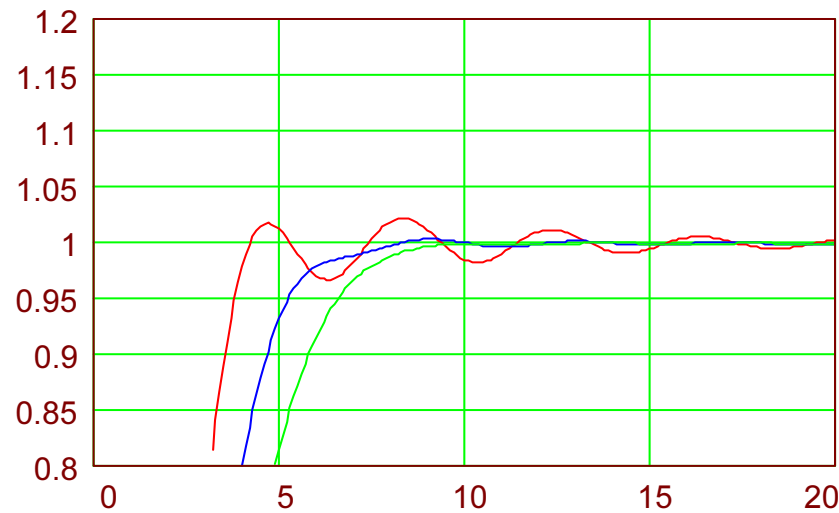
Unterminated line response

Risetime set to 4, 5 and 6 times transmission line delay

BLOWUP of vertical axis

$$ZS = 10 \quad RL = 1 \cdot 10^4$$

$$ZC = 65 \quad CL = 0$$



time, in units
of line delay

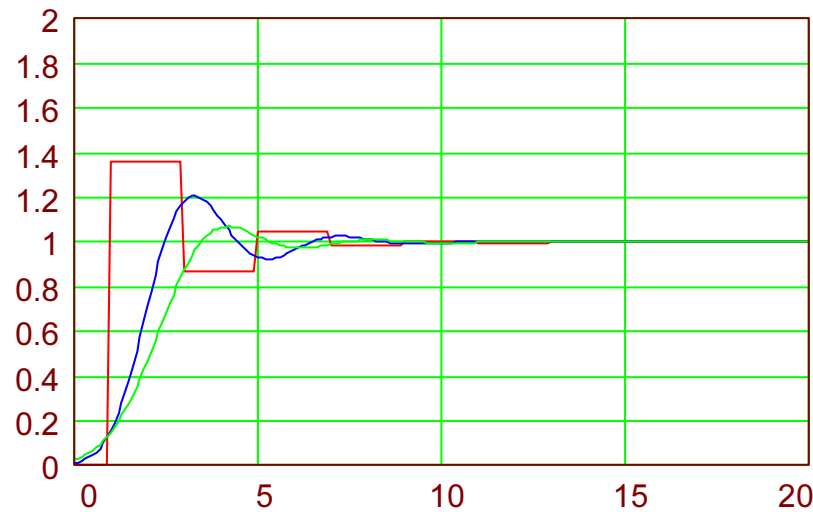
TTL/CMOS

Unterminated line response

Risetime set to 0, 2 and 3 times transmission line delay

$$Z_S = 30 \quad R_L = 1 \cdot 10^4$$

$$Z_C = 65 \quad C_L = 0$$



time, in units
of line delay

TTL/CMOS

Unterminated line response

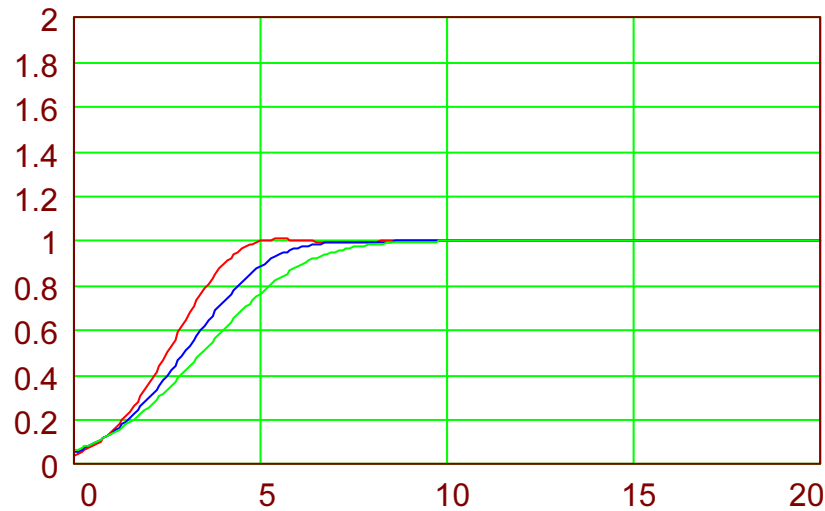
Risetime set to 4, 5 and 6 times transmission line delay

$$Z_S = 30$$

$$R_L = 1 \cdot 10^4$$

$$Z_C = 65$$

$$C_L = 0$$



time, in units
of line delay

TTL/CMOS

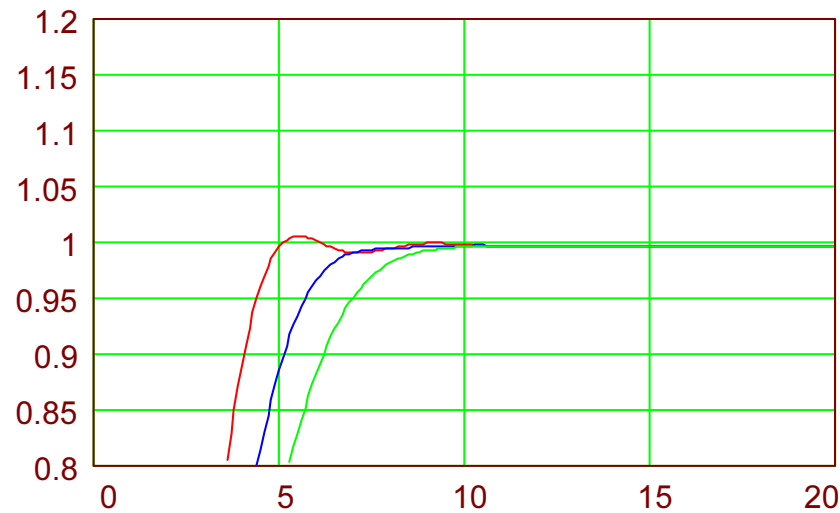
Unterminated line response

Risetime set to 4, 5 and 6 times transmission line delay

BLOWUP of vertical axis

$$Z_S = 30 \quad R_L = 1 \cdot 10^4$$

$$Z_C = 65 \quad C_L = 0$$



time, in units
of line delay

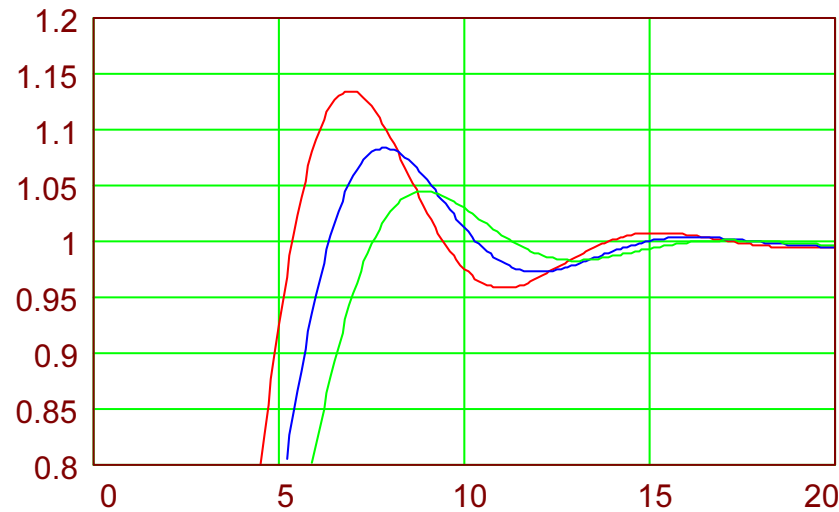
TTL/CMOS with Capacitive Load (20 pF)

Unterminated line response

Risetime set to 4, 5 and 6 times transmission line delay

BLOWUP of vertical axis

$$\begin{array}{ll} Z_S = 30 & R_L = 1 \cdot 10^4 \\ Z_C = 65 & C_L = 2 \cdot 10^{-11} \end{array}$$



time, in units
of line delay